DAVID TORCK

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Education

Texas A&M University, College Station TX

Nuclear Security Certification **Dec 2026**

MS in Engineering Technology – (Electro-Mechanical Control Systems) **Dec 2026**

BS in Multidisciplinary Engineering Technology – (Mechatronics Track) **Dec 2024**

Professional Experience

Amentum Propulsion Systems, Thrust Vector Control Intern - *NASA Marshall Space Flight Center* **May 2025 – Aug 2025**

- Engineered and integrated a single-axis active inertial loading simulator into existing TVC architecture, utilizing MATLAB for dynamic system modeling and LabVIEW for real-time data acquisition and hardware-in-the-loop control.
- Developed an active control system to replicate mission-representative loads, friction, and force profiles supporting the Mars Ascent Vehicle (MAV), and Space Launch System (SLS).

Autonomous Bio-photonic Robot for Phytopathology Detection Intern – *Texas A&M University* **Nov 2023 – Aug 2024**

- Designed a mobile robotic system to self-ruling position and trigger Raman spectroscopy scans across 96 corn stalks in a climate-controlled greenhouse, reducing operator intervention and experimental variability.
- Collected and quantitatively analyzed Raman spectrograph wavenumber data from Poaceae (hard stalk grass) samples to prompt controlled bio-molecular history.

Instrumentation and Control Electronics Intern - *Paragon Energy Solutions* **May 2023 – Aug 2023**

- Analyzed and tested electrical system architectures and logical control systems utilizing certified FPGA schematic capture direct from nuclear facilities including static timing analysis to check clock frequencies, power analysis to measure energy use, and logic simulation to verify behavior before hardware deployment.
- Executed certified soldering and drafting techniques to design universal testing equipment for multivariable logic boards.

Engineering Projects

Quantum Communication & Ghost Imaging Researcher – *Texas A&M University* **Jan 2025 – Dec 2026**

- Investigated integrations of quantum ghost imaging architectures with additively manufactured techniques and metamaterial lenses to generate practical, high-efficiency quantum-enabled imaging systems.
- Modeled physical foundations of quantum optics, leveraging photon correlations, two-path interference, and quantum measurement theory to establish a mathematical basis for an advanced imaging system.

Semi-Autonomous Mass Quality Control Robot (R.I.P.T.I.D.E.) – *Texas A&M University* **Jan 2024 – Dec 2024**

- Built a custom C++ intermediate controller to establish cross-protocol communication between a 6 degree of freedom robotic arm, end-of-arm tooling, and Creaform HandySCAN metrology software.
- Automated high-precision 3D inspections to a volumetric tolerance of ±0.001 inches, dynamically comparing digital twins against source CAD models to produce real-time deviation heat-maps.

Autonomous Drone for Reconnaissance and Sample Capture – *Texas A&M University* **Oct 2022 – Dec 2024**

- Co-engineered and deployed an autonomous, Mars-analogue robotic platform from scratch, integrating autonomous navigation and manipulation systems for in-situ soil sample extraction and biochemical life-detection analysis.

Leadership Background

Electronic Instrumentation Laboratory – *Texas A&M University* **Jan 2025 – Dec 2026**

- Prepared laboratory analog electronic systems and guided hands-on electrical instrumentation experiments while ensuring a safe, organized environment.
- Promoted student learning through clear technical communication, direct troubleshooting assistance, and one-on-one mentorship.

The Society of Mechatronics Engineering Technology – *Texas A&M University* **Jan 2022 – Dec 2024**

- President: Orchestrated strategic project timelines and cross-functional engineering events, collaborating with industry professionals to secure corporate sponsorships and technical mentorship.
- Treasurer: Managed end-to-end financial operations for S.O.M.T.E.C.H., overseeing accounts receivable/payable, debt mitigation, and delivering biweekly expenditure analysis to executive stakeholders.
- Science Team Lead: Science division's mission, establishing biochemical definitions and geological parameters required to justify potential biosignatures on the Martian surface.

Skills & Certifications

- Schematic capture using FPGA (Field-Programable Gate Array) in embedded systems. Utilized Verilog & VHDL programming.
- AutoCAD, Autodesk Fusion 360, Revit, Creo, and SolidWorks (TSTC occupational skills award in Architectural/Mechanical Drafting)
- PC J-STD-001 standards for circuit board repair and soldering.
- Proficient with Office 365 products (Excel, PowerPoint, & Word)
- LabView, MATLAB, Simulink
- Proficiency with Universal Robotics robotic arm platform and proprietary PolyScope software.
- Creaform Portable 3D Scanner Training: HandyScan3D, VXmodel, VXinspect, and I/O Module